Practical-5

Problem Definition:

- Design and simulate a 4-bit dual-mode counter using JK Flip- Flops that can count up or down based on the MODE control input.

The counter must include:
● An asynchronous CLEAR input to reset the counter to 0000.
● A synchronous LOAD feature to load any 4-bit preset value on command.
The design should include excitation tables, Boolean expressions, and optimized logic equations for all J and K inputs, verified through truth tables and simulation.

Requirements / Specifications:

●Use JK Flip-Flops only.

●Implement MODE control for Up/Down counting.

●Include asynchronous CLEAR to reset the counter.

●Include synchronous LOAD to preset the counter with a 4 bit input value.

●Derive excitation table and Boolean expressions for J and K inputs.

●Optimize logic using K-map simplification.

●Verify the circuit operation using simulation software (Logisim / Proteus / Multisim).

Tools/Technology To Be Used:

●Simulation Software: Logisim Evolution / Proteus / Multisim / Tinkercad Circuits

●Components: JK Flip-Flops, Logic Gates, Clock Pulse Generator

●Verification: Truth table and waveform simulation

Aim:

To design and simulate a 4-bit dual-mode counter using JK Flip-Flops that can count in both up and down directions based on a MODE control input, with features of asynchronous CLEAR for reset and synchronous LOAD for presetting values, and to verify its operation using truth tables and simulation tools.

J-K flip flop:

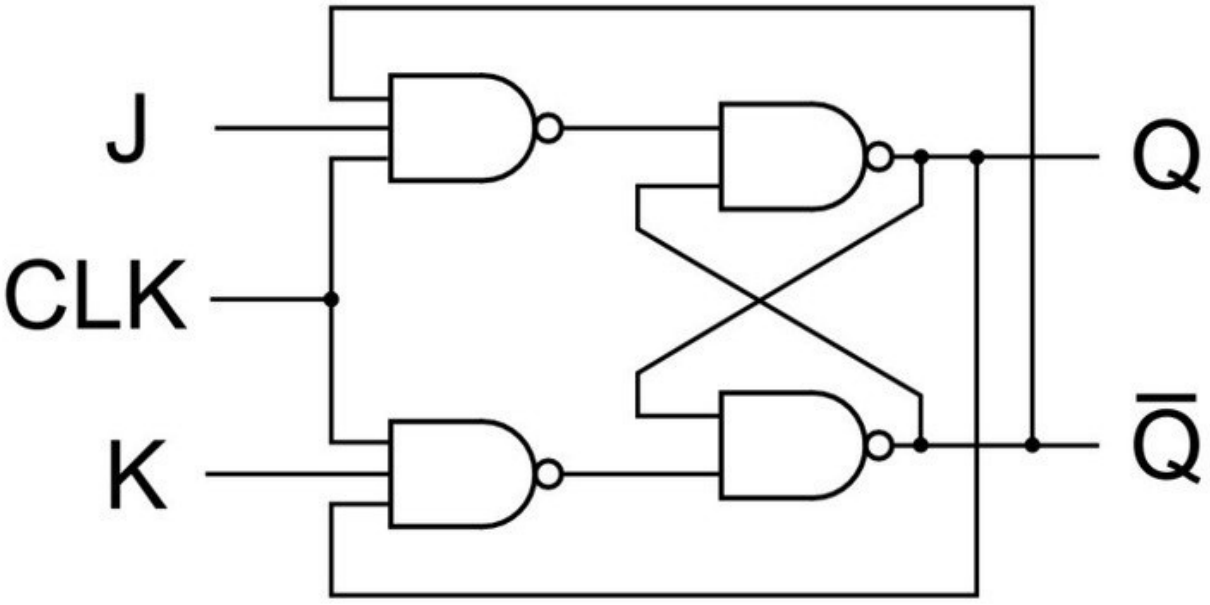
Truth Table:

J	K	Qn	Qn+1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation Table:

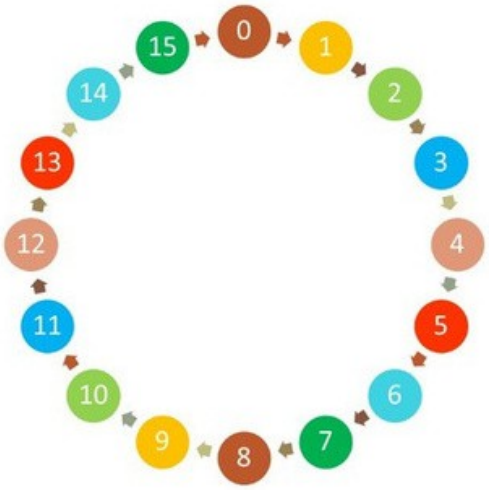
Q	Qn+1	J	J
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Circuit diagram:



4-bit binary up counter:

State diagram:

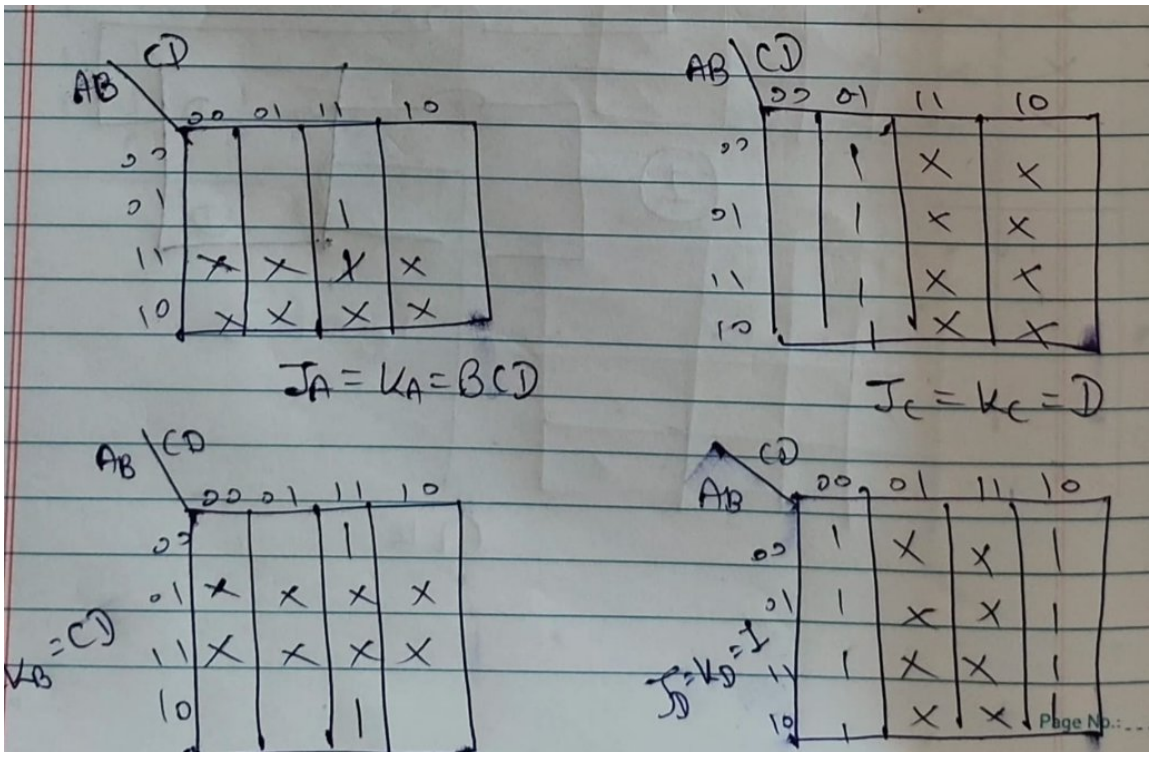


State table:

A_0	B	C	D	A_+	B_+	C_+	D_+	$J_A K_A$	$J_B K_B$	$J_C K_C$	$J_D K_D$
0	0	0	0	0	0	0	1	0 x	0 x	0 x	1 x
0	0	0	1	0	0	1	0	0 x	0 x	1 x	x 1
0	0	1	0	0	0	1	1	0 x	0 x	x 0	1 x
0	0	1	1	0	1	0	0	0 x	1 x	x 1	x 1
0	1	0	0	0	1	0	1	0 x	x 0	0 x	1 x
0	1	0	1	0	1	1	0	0 x	x 0	1 x	x 1
0	1	1	0	0	1	1	1	0 x	x 0	x 0	1 x
0	1	1	1	1	0	0	0	1 x	x 1	x 1	x 1
1	0	0	0	1	0	0	1	x 0	0 x	0 x	1 x
1	0	0	1	1	0	1	0	x 0	0 x	1 x	x 1
1	0	1	0	1	0	1	1	x 0	0 x	x 0	1 x
1	0	1	1	1	1	0	0	x 0	1 x	x 1	x 1
1	1	0	0	1	1	0	1	x 0	x 0	0 x	1 x
1	1	0	1	1	1	1	0	x 0	x 0	1 x	x 1
1	1	1	0	1	1	1	1	x 0	x 0	x 0	1 x
1	1	1	1	0	0	0	0	x 1	x 1	x 1	x 1

K -map:

25DCE014



Equation:

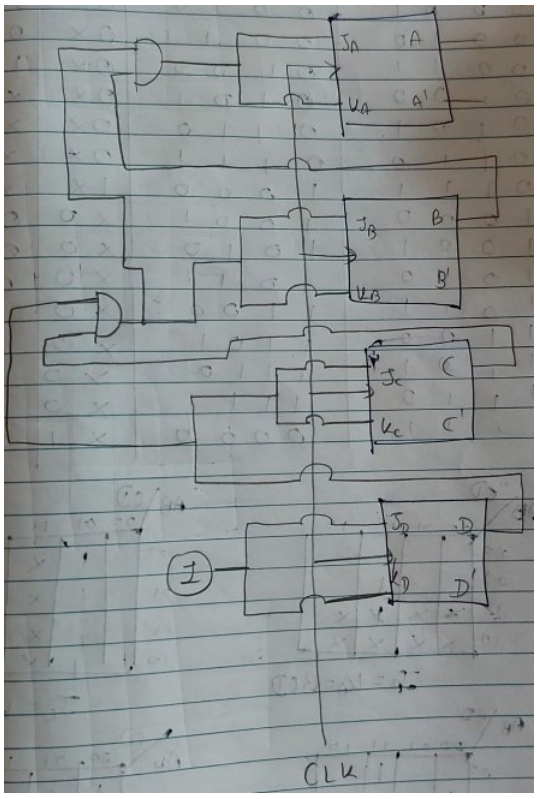
For Flip-Flop A: $J^A = K^A = B \cdot C \cdot D$

For Flip-Flop B: $JB = K^B = C \cdot D$

For Flip-Flop C: $J^C = K^C = D$

For Flip-Flop D: $JD = K^D = 1$

Circuit:



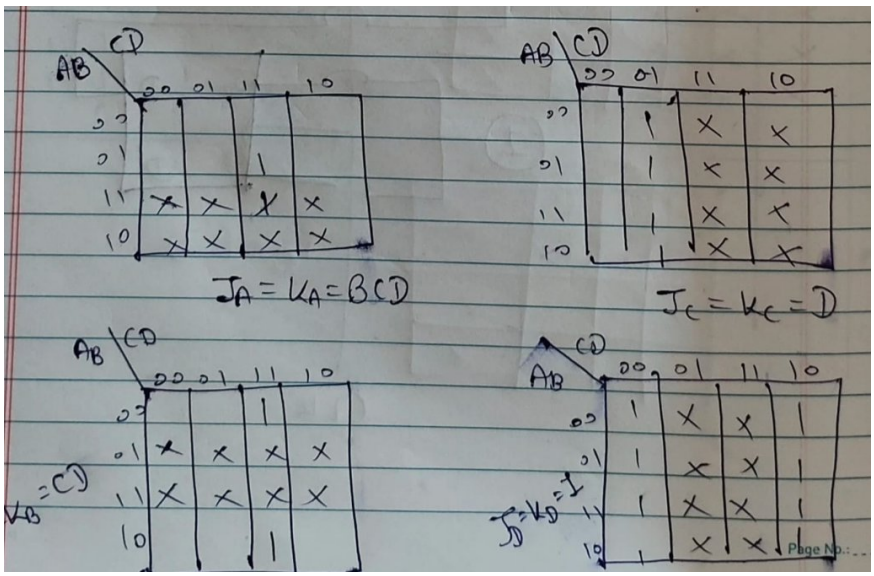
Down counter:

State Table:

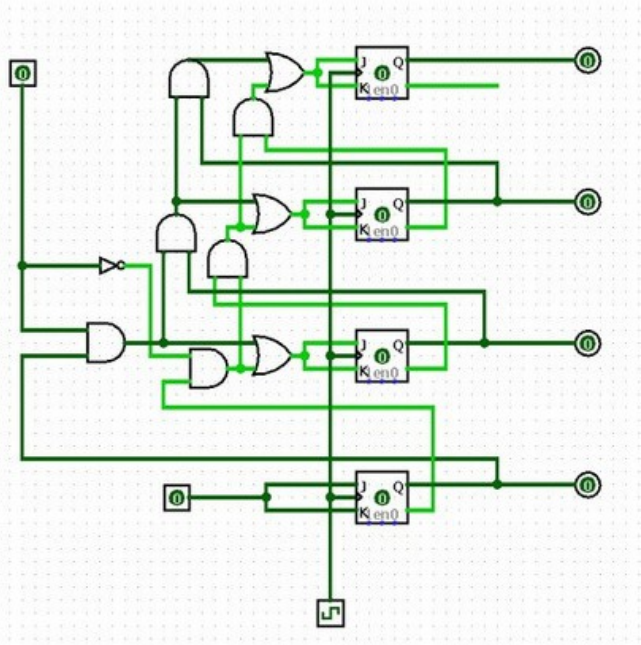
STATE	A	B	C	D	A+	B+	C+	D+	JA	KA	JB	KB	JC	KC	JD	KD
0	0	0	0	0	1	1	1	1	1	X	1	X	1	X	1	X
1	0	0	0	1	0	0	0	0	0	X	0	X	0	X	X	1
2	0	0	1	0	0	0	0	1	0	X	0	X	X	1	1	X
3	0	0	1	1	0	0	1	0	0	X	0	X	X	0	X	1
4	0	1	0	0	0	0	1	1	0	X	X	1	1	X	1	X
5	0	1	0	1	0	1	0	0	0	X	X	0	0	X	X	1

STATE	A	B	C	D	A +	B +	C +	D +	J A	K A	J B	K B	J C	K C	J D	K D
6	0	1	1	0	0	1	0	1	0	X	X	0	X	1	1	X
7	0	1	1	1	0	1	1	0	0	X	X	0	X	0	X	1
8	1	0	0	0	0	1	1	1	X	1	1	X	1	X	1	X
9	1	0	0	1	1	0	0	0	X	0	0	X	0	X	X	1
10	1	0	1	0	1	0	0	1	X	0	0	X	X	1	1	X
11	1	0	1	1	1	0	1	0	X	0	0	X	X	0	X	1
12	1	1	0	0	1	0	1	1	X	0	X	1	1	X	1	X
13	1	1	0	1	1	1	0	0	X	0	X	0	0	X	X	1
14	1	1	1	0	1	1	0	1	X	0	X	0	X	1	1	X
15	1	1	1	1	1	1	1	0	X	0	X	0	X	0	X	1

K-map & Equation:

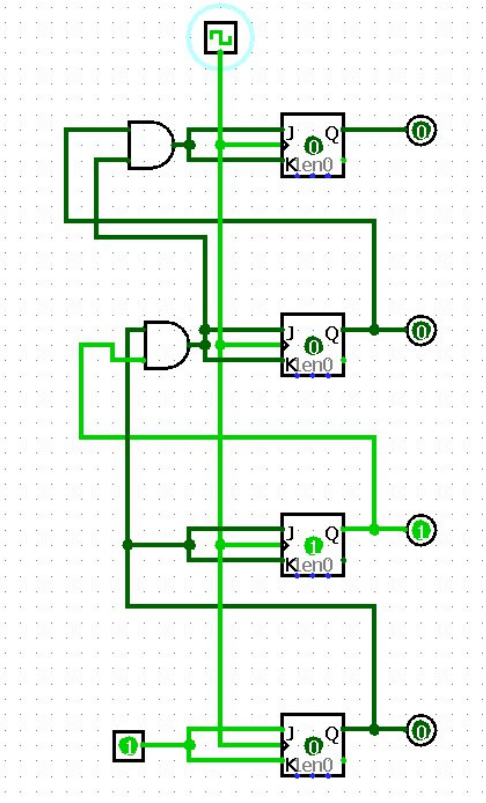


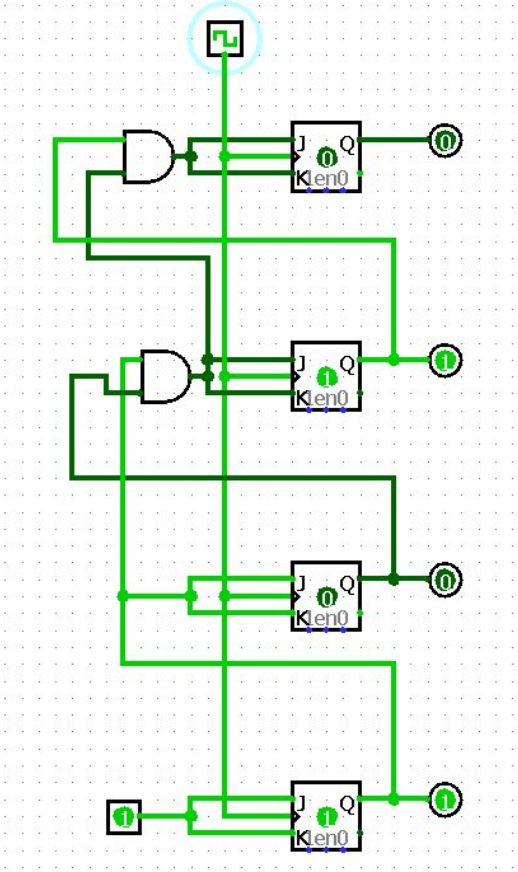
4 Bit Combine circuit:

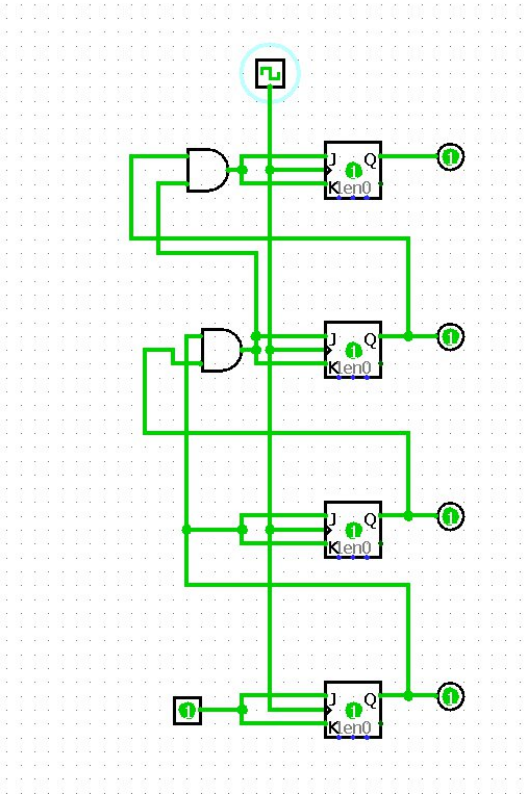


Screen shots:

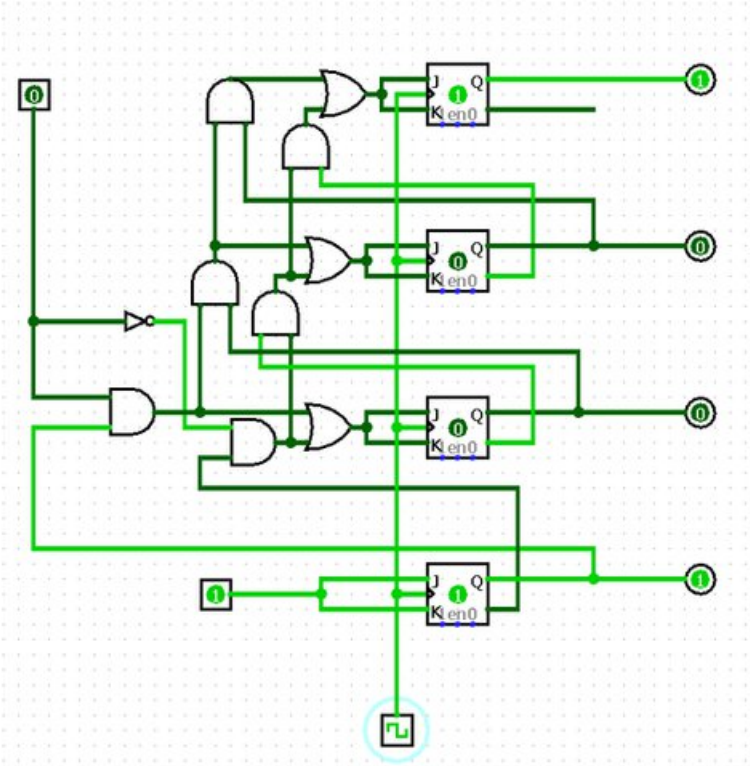
Up-counter:

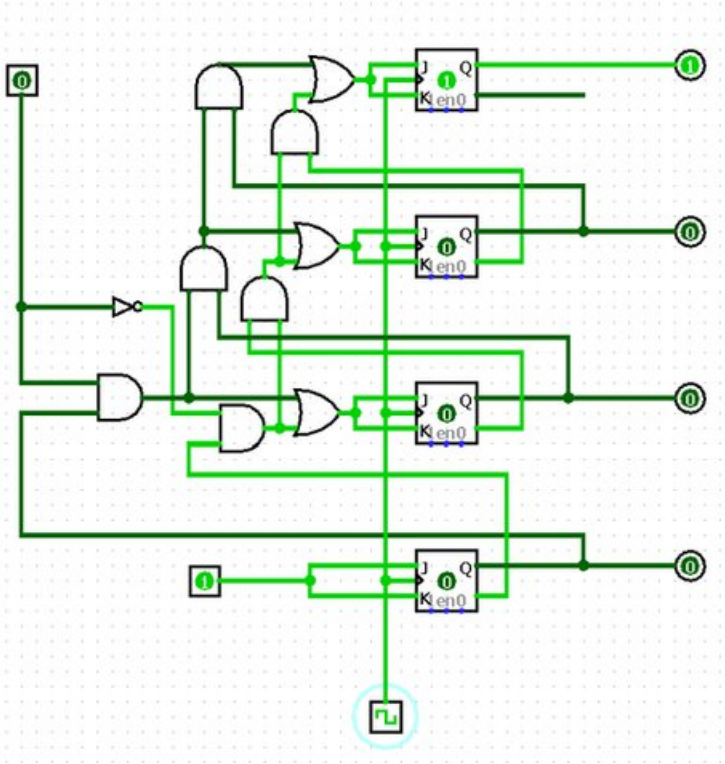


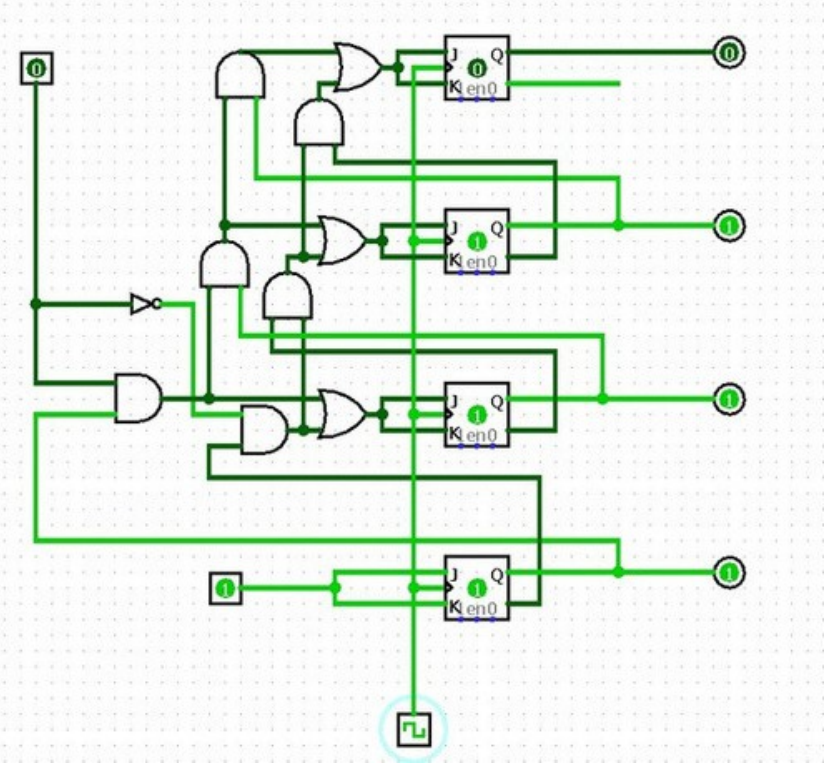

J
Q
K
en
0
1


J
Q
K
en
0
1

Combine (down counter):







Conclusion:

The 4-bit dual-mode counter using JK flip-flops was successfully designed and simulated. It correctly performs both up and down counting based on the MODE input. The asynchronous CLEAR resets the counter to 0000, and the synchronous LOAD allows presetting any value. The excitation tables and Boolean expressions were simplified using K-maps, and simulation verified proper operation of the circuit.